

GRIDER FIELD, AR, USA

WMO: 723417

Lat: 34.180N

Lon: 91.934W

Elev: 207

StdP: 14.59

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93988

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.5	25.2	8.6	8.6	26.5	13.2	10.9	29.6	22.8	54.7	20.5	50.2	8.1	10	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	97.2	77.6	94.9	77.3	92.7	76.8	80.4	91.5	79.5	90.6	78.5	89.1	6.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	143.8	85.4	76.6	139.6	84.7	75.3	133.7	83.7	44.1	91.6	43.0	90.5	42.1	89.6	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.9	17.8	16.1	DB	16.1	100.0	4.6	3.3	12.8	102.4	10.1	104.3	7.5	106.2	4.2	108.6
			WB	14.0	82.8	4.3	1.7	10.9	84.1	8.4	85.1	6.0	86.0	2.9	87.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	43.2	47.0	54.8	63.8	72.1	79.5	82.0	81.4	75.2	64.4	53.2	45.4
	DBStd	15.95	10.68	10.35	9.59	7.64	6.33	4.23	4.03	4.35	5.91	8.02	8.78	9.53
	HDD50	772	265	166	61	5	0	0	0	0	0	3	65	207
	HDD65	2746	678	509	337	111	18	0	0	0	5	113	366	609
	CDD50	5728	56	81	208	418	685	884	993	973	757	448	160	65
	CDD65	2226	3	4	19	74	238	434	528	508	312	93	11	2
	CDH74	22424	9	20	134	589	2010	4493	5872	5431	2921	876	64	5
	CDH80	9989	0	1	14	108	676	2049	2875	2719	1273	269	5	0
Wind	WSAvg	6.7	8.0	8.3	8.6	8.4	7.1	5.8	4.9	4.5	5.2	6.0	6.8	7.5
Precipitation	PrecAvg	50.9	4.1	4.3	5.2	4.8	5.0	3.5	3.8	3.3	3.5	4.2	4.4	5.2
	PrecMax	72.5	8.3	10.6	13.7	16.3	13.0	8.7	12.9	8.0	10.3	14.1	13.5	15.0
	PrecMin	32.8	0.5	0.7	1.0	1.0	0.8	0.1	0.2	0.0	0.1	0.0	0.7	0.7
	PrecStd	10.2	2.1	2.2	2.5	3.4	3.0	2.2	2.3	2.0	2.1	3.0	2.7	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	77.0	82.0	86.4	91.4	97.2	99.6	101.6	97.2	89.6	80.3	74.1
		MCWB	64.1	64.8	65.8	69.6	73.8	76.4	78.9	77.9	73.0	70.5	66.7	65.7
	2%	DB	69.7	72.3	78.1	83.1	88.9	94.0	96.8	97.5	93.4	86.4	75.2	69.7
		MCWB	62.4	62.5	64.5	67.7	72.5	76.4	78.5	78.2	74.5	70.0	64.1	63.2
	5%	DB	64.9	68.2	74.0	80.7	86.6	91.8	94.5	94.8	90.6	82.8	72.0	65.3
		MCWB	59.5	60.7	62.4	66.7	71.9	75.9	78.2	77.5	73.9	68.1	63.6	60.3
	10%	DB	60.9	64.1	70.3	77.4	84.0	90.1	92.2	92.2	87.8	79.3	68.1	61.3
		MCWB	54.7	57.5	60.5	65.2	71.3	75.3	77.9	77.0	72.9	66.5	61.1	56.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.9	67.5	69.1	73.4	77.4	80.5	82.2	81.9	79.0	75.5	70.3	67.8
		MCDB	71.8	73.3	77.8	81.9	86.6	90.8	92.9	93.1	89.9	84.0	75.6	72.5
	2%	WB	64.1	65.2	66.8	71.1	75.4	78.8	80.7	80.3	77.3	73.5	67.6	64.7
		MCDB	67.9	70.2	74.9	79.4	84.2	89.1	91.7	91.9	87.8	81.5	72.8	68.2
	5%	WB	60.9	62.4	64.7	69.4	74.2	77.8	79.8	79.2	76.1	71.5	65.1	61.6
		MCDB	64.5	67.1	71.7	76.9	83.0	88.0	90.7	90.8	85.5	78.2	70.3	65.1
	10%	WB	55.6	58.4	62.3	67.6	72.9	76.9	78.8	78.2	74.9	69.3	62.2	57.3
		MCDB	59.4	63.2	68.5	74.6	81.4	86.6	89.3	89.2	83.5	75.8	66.9	60.6
Mean Daily Temperature Range	5% DB	MDBR	17.2	18.0	19.3	20.3	18.8	18.6	18.4	19.6	21.2	22.2	19.7	16.8
		MCDBR	22.6	24.4	23.5	22.8	21.1	20.5	20.9	22.9	24.4	26.4	23.8	21.6
		MCWBR	16.5	16.4	13.1	10.9	8.4	7.2	7.3	7.6	8.5	11.8	14.9	16.2
	5% WB	MCDBR	19.6	21.2	20.3	18.8	18.2	18.3	18.9	20.5	19.8	19.8	20.0	19.3
		MCWBR	16.3	16.2	13.3	11.0	8.5	7.5	7.6	7.9	7.9	11.2	14.9	16.0
Clear-Sky Solar Irradiance	taub	0.322	0.332	0.361	0.402	0.434	0.463	0.492	0.486	0.419	0.361	0.338	0.318	
	taud	2.507	2.470	2.413	2.296	2.260	2.201	2.155	2.158	2.334	2.471	2.492	2.527	
	Ebn at Noon	282	291	290	281	272	263	254	253	267	277	273	277	
	Edh at Noon	27	31	35	42	43	46	48	47	38	30	27	25	
All-Sky Solar Radiation	RadAvg	802	949	1331	1673	1844	2059	1991	1834	1574	1239	888	709	
	RadStd	72	147	120	113	134	151	110	143	124	165	100	77	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page